MILAN MATHEW

mathewmilan86@gmail.com | +91 8089027452 | Portfolio: reachmilan.com
GitHub: github.com/MilanMathew86 | LinkedIn: linkedin.com/in/milan-mathew-19216b326

EDUCATION

B.Tech, Computer Science and Engineering | Chinmaya Vishwa Vidyapeeth, Ernakulam, Kerala 2024 – 2028
CGPA: **9.72 / 10**
Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems (NPTEL), Theory of Computation (NPTEL)

EXPERIENCE

Summer Research Intern, Applied Machine Learning in Speech | National Institute of Technology (NIT) Goa Summer 2026
Mentor: Dr. Veena Thenkanidiyoor, Associate Professor and Head, Department of CSE

- Built a spoken language identification system for 8 Indian languages using the IIT Mandi ReadSpeech dataset, using MFCC feature extraction and feature standardization.
- Benchmarked 8 classifiers (Naive Bayes, Perceptron, SVM, Random Forest, KNN, MLP, XGBoost, tuned SVM) with 5-fold cross-validation and confusion-matrix analysis; tuned SVM achieved the best test accuracy of **91.99%**.
- Deployed the best model in a Flask web app that predicts the language of an input audio recording.

Web Development Intern | Tisser Technologies June 2025 (1 month)

- Built responsive web interfaces and interactive forms using HTML, CSS and JavaScript (DOM manipulation); debugged and refactored code to improve structure and maintainability.

PROJECTS

CVV Hostel E-Outpass System | Flutter, PHP, MySQL Completed 2026

- Developed a full-stack application to replace the college's manual outpass process (paper, email and WhatsApp); submitted to the college administration for adoption as the official outpass system.
- Implemented parent OTP login, role-based approval routing to teachers and wardens based on academic-calendar rules, and QR e-outpasses scanned at the main gate for exit and return.

CoralSense AI: AI-Integrated Drifting Buoy for Coral Reef Health Assessment | Oceanic AI Lab (college-funded) Ongoing

- Co-developing (team of two) a low-cost, current-drifting buoy that combines underwater imaging, temperature, pH and salinity sensing, and surface GPS to produce geo-referenced reef-health records.
- Building the AI pipeline: underwater image enhancement (colour, contrast, visibility correction) and U-Net semantic segmentation classifying coral as Healthy, Bleached or Dead, using the Coralscapes dataset (2,075 expert-annotated images); software stage nearing completion, hardware integration next.

Crop Yield Prediction System | Python, Scikit-learn, XGBoost, Flask

- Built an end-to-end ML pipeline on a 1M+ row dataset; trained Linear Regression and XGBRFRegressor models ($R^2 \approx 0.91$) and deployed the model as a Flask web application.

PUBLICATIONS AND PRESENTATIONS

- **Uncertainty-Aware, Explainable AutoML for Crop Yield Prediction: A Conformal Prediction Approach** (First Author). Accepted and presented at Big Data and AI 2026 (BDA 2026), BITS Pilani, September 2026.
- **Uncertainty-Aware Distributed Clustering of Streaming Data** (First Author). Under review, ICITIIT 2027, IIIT Kottayam.
- **Algorithmic Myopia in Platform Ecosystems: Exploitation vs. Entry in Gig-Economy Labor Markets** (Co-author). Accepted after double-blind review for poster presentation, India Strategy Conference (ISC) 2026, IIM Bangalore, December 2026.
- **CoralSense AI: AI-Integrated Drifting Buoy Platform for Coral Reef Health Assessment** (Co-author). Abstract, oral presentation, OCEANAI-2026, Sathyabama Research Park, Chennai, September 2026.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, PHP, Dart, SQL
Web and Mobile: Flutter, Flask, HTML, CSS
Machine Learning: Scikit-learn, XGBoost, NumPy, Pandas, Matplotlib; classification, regression, cross-validation, conformal prediction, MFCC speech features, U-Net semantic segmentation
Databases and Tools: MySQL, Git

ACHIEVEMENTS

- Second Prize, Smart India Hackathon (SIH) 2025 Internal Round, for *HydroNet* (AI + IoT hydroponics concept).
- Second Prize, Smart India Hackathon (SIH) 2024 Internal Round, for *Ayushvatika: Virtual Herbal Garden*.
- First Prize, Catalyst Idea Pitch (inter-college competition), for *HydroNet*.

POSITIONS OF RESPONSIBILITY

Documentation Lead, IEEE Oceanic Engineering Society (OES), CVV Student Branch | Documentation Lead, IEDC, CVV